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AIM: Implement Holdout Validation, K-Fold, Bootstrap

Dataset: BankLoan Dataset

```
import numpy as np
import pandas as pd
import kagglehub

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.utils import resample
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
# ----- Download Dataset from Kaggle -----
path = kagglehub.dataset_download("vikramamin/bank-loan-approval-lr-dt-rf-and-auc")

# Load dataset
data = pd.read_csv(path + "/bankloan.csv")

print("Dataset Preview:")
print(data.head())
```

```
Downloading to /root/.cache/kagglehub/datasets/vikramamin/bank-loan-approval-lr-dt-rf-and-auc/1.archive...
100%|██████████| 60.0k/60.0k [00:00<00:00, 1.16MB/s]Extracting files...
```

Dataset Preview:

	ID	Age	Experience	Income	ZIP.Code	Family	CAvg	Education	Mortgage	\
0	1	25	1	49	91107	4	1.6	1	0	
1	2	45	19	34	90089	3	1.5	1	0	
2	3	39	15	11	94720	1	1.0	1	0	
3	4	35	9	100	94112	1	2.7	2	0	
4	5	35	8	45	91330	4	1.0	2	0	

	Personal.Loan	Securities.Account	CD.Account	Online	CreditCard
0	0		1	0	0
1	0		1	0	0
2	0		0	0	0
3	0		0	0	0
4	0		0	0	1

```
# ----- Data Preprocessing -----
```

```
# Remove unnecessary columns and separate features and target
X = data.drop(["ID", "ZIP.Code", "Personal.Loan"], axis=1)
y = data["Personal.Loan"]

# Feature scaling for SVM
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

```
# ----- Holdout Validation -----
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

svm_model = SVC(kernel='poly', degree=3, coef0=1, C=1.0)

svm_model.fit(X_train, y_train)

y_pred = svm_model.predict(X_test)
```

```

print("\nHoldout Evaluation:")
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))

```

```

Holdout Evaluation:
Accuracy: 0.985
Confusion Matrix:
[[893  2]
 [ 13 92]]
Classification Report:

```

	precision	recall	f1-score	support
0	0.99	1.00	0.99	895
1	0.98	0.88	0.92	105
accuracy			0.98	1000
macro avg	0.98	0.94	0.96	1000
weighted avg	0.98	0.98	0.98	1000

```

# ----- Bootstrap Sampling -----

X_boot, y_boot = resample(X, y, replace=True, n_samples=len(X), random_state=1)

X_boot_train, X_boot_test, y_boot_train, y_boot_test = train_test_split(
    X_boot, y_boot, test_size=0.2, random_state=42
)

svm_model_boot = SVC(kernel='poly', degree=3, coef0=1, C=1.0)

svm_model_boot.fit(X_boot_train, y_boot_train)

y_boot_pred = svm_model_boot.predict(X_boot_test)

print("\nBootstrap Evaluation:")
print("Accuracy:", accuracy_score(y_boot_test, y_boot_pred))
print("Confusion Matrix:\n", confusion_matrix(y_boot_test, y_boot_pred))
print("Classification Report:\n", classification_report(y_boot_test, y_boot_pred))

```

```

Bootstrap Evaluation:
Accuracy: 0.987
Confusion Matrix:
[[909  2]
 [ 11 78]]
Classification Report:

```

	precision	recall	f1-score	support
0	0.99	1.00	0.99	911
1	0.97	0.88	0.92	89
accuracy			0.99	1000
macro avg	0.98	0.94	0.96	1000
weighted avg	0.99	0.99	0.99	1000

```

# ----- K-Fold Cross Validation -----

cv_scores = cross_val_score(
    SVC(kernel='poly', degree=3, coef0=1, C=1.0),
    X,
    y,
    cv=10
)

print("\nK-Fold Cross Validation:")
print("Cross Validation Scores:", cv_scores)
print("Average CV Accuracy:", np.mean(cv_scores))

```

```
K-Fold Cross Validation:  
Cross Validation Scores: [0.98  0.982 0.982 0.974 0.978 0.982 0.978 0.99  0.986 0.976]  
Average CV Accuracy: 0.9808
```

```
# ----- Final Accuracy Comparison -----  
  
holdout_acc = accuracy_score(y_test, y_pred)  
bootstrap_acc = accuracy_score(y_boot_test, y_boot_pred)  
kfold_acc = np.mean(cv_scores)  
  
print("\nAccuracy Comparison:")  
print("Holdout Accuracy:", holdout_acc)  
print("Bootstrap Accuracy:", bootstrap_acc)  
print("K-Fold Accuracy:", kfold_acc)
```

```
Accuracy Comparison:  
Holdout Accuracy: 0.985  
Bootstrap Accuracy: 0.987  
K-Fold Accuracy: 0.9808
```

Inference

The SVM model performs very well on the bank loan dataset, achieving high accuracy in all evaluation methods. Bootstrap validation produced the highest accuracy of 0.987, indicating that the model performs slightly better when trained on resampled data. Holdout validation also shows strong performance with an accuracy of 0.985, confirming that the model generalizes well on unseen test data. The K-Fold cross validation accuracy of 0.9808 is slightly lower but still very close to the other results, suggesting consistent performance across different folds of the dataset. Overall, the small difference between the three accuracies indicates that the model is stable, reliable, and not significantly affected by overfitting.